

EX:NO:31

PROGRAM:

```
def merge_sort(arr):
    if len(arr) > 1:
        mid = len(arr)//2
        left = arr[:mid]
        right = arr[mid:]

        merge_sort(left)
        merge_sort(right)

    i=j=k=0
    while i < len(left) and j < len(right):
        if left[i] < right[j]:
            arr[k] = left[i]
            i+=1
        else:
            arr[k] = right[j]
            j+=1
        k+=1

    while i < len(left):
        arr[k] = left[i]
        i+=1
        k+=1

    while j < len(right):
```

```
arr[k] = right[j]
```

```
j+=1
```

```
k+=1
```

```
arr = [31,23,35,27,11,21,15,28]
```

```
merge_sort(arr)
```

```
print("Sorted:", arr)
```

Output

```
Sorted: [11, 15, 21, 23, 27, 28, 31, 35]
```

```
=== Code Execution Successful ===
```